Development of a Remote Sensing System for Real-Time Detection of Military Threats

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***Abstract*—To strengthen the national defense system, advances in remote sensing technology are essential, especially to detect military threats that could threaten sovereignty. These threats can be physical or non-physical, such as cyber-attacks that can damage vital infrastructure. Therefore, effective border monitoring is critical to maintaining the security and stability of the country. Although remote sensing technology is developing rapidly, several issues hinder its implementation, such as inadequate infrastructure and lack of employee training. Previous research has not fully studied how various advanced technologies can be incorporated into Indonesia's border area defense system that is responsive and operates in real time. The purpose of this study is to see how well remote sensing systems detect military threats and evaluate the problems that arise when implementing them. In addition, this study aims to provide strategic suggestions that will help improve national defense. This research utilizes a literature review of relevant articles and analyzes information to gain a further understanding of advances in remote sensing technology and its applications in defense. Studies show that technologies such as radar, CCTV, and artificial intelligence (AI) can improve threat detection in border areas. AI-based radars enable faster identification and classification of threats, and cloud-based CCTV improves real-time monitoring. The findings from this study suggest that the development and utilization of advanced remote sensing technologies can improve Indonesia's defense system. By addressing infrastructure and personnel training issues, the country can more effectively safeguard its sovereignty and stability by improving responses to military and non-military threats.**

Keywords—border surveillance, defense systems, remote sensing, threats

I. Introduction

Technological advancements demand continuous adaptation to face various forms of threats, both physical and non-physical. One serious physical threat that requires attention is military threats, particularly in the form of border violations. Such violations can directly impact a country's sovereignty, necessitating a defense system capable of real-time monitoring and responsive action against various threats. In this context, remote sensing technology plays a crucial role in building effective and efficient defense systems [1]. Remote sensing systems have rapidly developed and are increasingly capable of detecting military threats by utilizing various advanced devices such as radar, infrared cameras, and Closed Circuit Television (CCTV). This technology is integrated into perimeter intrusion detection systems (PIDS) and physical intrusion detection systems (IDS) to provide comprehensive surveillance at national borders. The advantage of this technology lies in its ability to automatically detect threats, even in challenging environmental conditions such as bad weather or nighttime, thereby enhancing early threat prevention effectiveness [2]. The advancement of integrated remote sensing technology within defense systems allows for real-time and responsive monitoring of physical threats, thus improving the effectiveness of national defense systems in preventing early threats.

A defense system that can be monitored in real-time has several strategic objectives, such as deterrence, detection, defense, delay, and denial of access. All these objectives are designed to enhance the quality of defense, especially in Indonesia's border areas, which frequently face physical military threats from outside. According to [1], early threat detection through remote sensing systems enables defense forces to respond quickly and accurately, thereby reducing the risk of violations and escalating conflicts at the borders. However, despite remote sensing offering effective solutions for detecting military threats, its implementation still faces various challenges. One major challenge is the need for adequate technological infrastructure, including the high costs associated with building and maintaining these systems. Additionally, the success of implementing this technology heavily depends on the capability and willingness of leaders to make strategic decisions regarding technology adoption [3]. The existing human resource conditions at the borders also play a crucial role that must be considered. Personnel stationed at the borders require specialized training to operate and maximize the use of this technology so that the defense system can function optimally and respond effectively [4], [5]. Remote sensing technology using radar and advanced sensor integration systems significantly enhances threat detection capabilities in hard-to-reach border areas, especially in regions with complex topography and challenging weather conditions [1]. Furthermore, other studies indicate that using CCTV connected to AI-based video management systems can reduce the risk of detection errors, which can expedite responses to emerging threats [6]. The combination of remote sensing and artificial intelligence not only improves detection but also enables faster and more accurate decision-making by security personnel.

A real-time defense system at Indonesia's borders aims to prevent, detect, defeat, delay, and deny military threats. However, its success depends heavily on technological infrastructure, strategic decisions by leaders, and specialized training for personnel at the borders. The Indonesian government, through the Telecommunications and Information Accessibility Agency (BAKTI) under the Ministry of Communication and Digital, has launched a program to develop supporting communication infrastructure such as Internet Access, Base Transceiver Stations (BTS), the PALAPA RING project, and the Satellite of the Republic of Indonesia (SATRIA). As illustrated in Fig. 1, the map shows the distribution of Internet Access across Indonesia [7]. Out of a total of 14,360 Internet Access locations, 9,955 locations are already connected, while 4,405 locations are still under evaluation. The distribution of Base Transceiver Stations (BTS), as of April 9, 2023, a total of 4,341 4G BTS are

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operational, while the remaining 1,277 BTS are still under construction from the target of 5,628 4G BTS installations. Meanwhile, the total of 1,682 Universal Service Obligation (USO) BTS have been established across Indonesia. USO BTS are those funded by the Universal Service Obligation (USO) program.

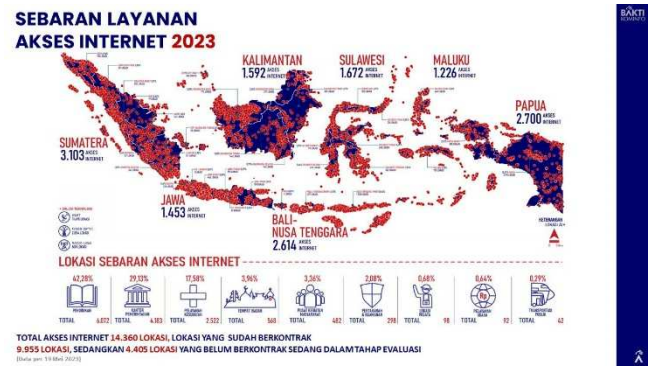

Fig. 1. Distribution Map of Internet Access Across Indonesia

The strengthening of communication networks undertaken by the Indonesian government is not limited to Internet Access and BTS. It also includes the PALAPA RING program. PALAPA RING is a national fiber optic network aimed at equitable broadband access throughout Indonesia. Fig. 2 provides a map of the distribution of the PALAPA RING network [7]. According [8] the total length of the marine cable communication system fiber optic network has reached 118,257.54 km, while terrestrial fiber optic spans 716,935.55 km, covering 34 provinces, 503 cities/regencies, 5,564 sub-districts, and 47,483 villages/urban villages. In addition to the three programs mentioned, Indonesia has also developed a satellite named Satellite of the Republic of Indonesia (SATRIA). This satellite was constructed to provide internet access, particularly in the Outermost, Disadvantaged, and Frontier areas and along the borders of the Unitary State of the Republic of Indonesia [7].

Fig. 2. Distribution Map of PALAPA RING Network Across Indonesia

The essential part for implementing and scaling of real-time defense systems across Indonesia, particularly in border regions, requires careful consideration of the country's diverse infrastructure capabilities, economic conditions, and security needs. By leveraging existing communication infrastructure like the PALAPA RING network, BTS installations, and the SATRIA satellite, the systems can be deployed effectively, even in areas with limited resources. A hybrid approach combining terrestrial and satellite technologies, along with strategic investments in infrastructure and innovative funding models, will ensure that Indonesia's border security system remains robust and adaptable to the country's varied geographic and economic conditions. For example, the government could explore such as public-private partnerships and crowdsource funding models. These efforts will not only enhance national security but also support Indonesia's broader goals of digital inclusion and sustainable development.

II. Methods

The methodology employed in this study includes a literature review of relevant journals, as well as an analysis to understand the advancements in remote sensing technology and its applications in defense contexts.

A. Research Framework

The research process begins by identifying various types of military threats that may be encountered to determine the needs in the development of sensing systems. After understanding the types of threats, the study focuses on exploring emerging sensing technologies that can be used to effectively detect and respond to threats. From the understanding of existing sensing technologies, this research then establishes the objectives for their application in the context of military threats, including early detection, real-time monitoring, and the ability to provide rapid responses to potential threats. By knowing the objectives of developing remote sensing technology to counter threats, this study can recommend defense systems that can be utilized based on an analysis of the effectiveness of existing technologies and how they can be integrated into national defense strategies.

Finally, this research also aims to identify the challenges faced in implementing remote sensing technology within defense contexts, including challenges related to infrastructure, investment costs, and human resource training, as well as discuss ways to overcome these challenges so that the sensing systems can function optimally. With this systematic methodological approach, it is hoped that this study can provide a comprehensive understanding of the development of real-time sensing systems for detecting military threats and make a significant contribution to enhancing the effectiveness of national defense strategies in Indonesia. Fig. 3 shows the research framework as a tool to facilitate the understanding of this study.

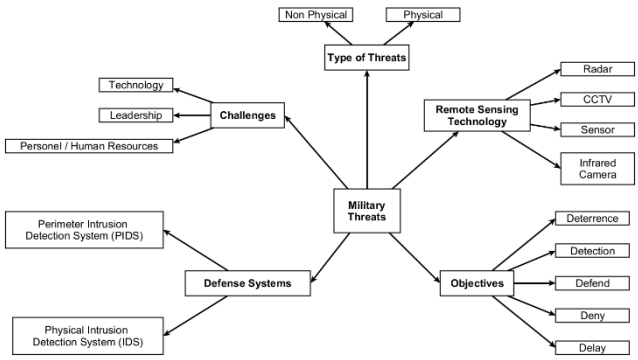

Fig. 3. Research Framework

B. The Effectiveness of Remote Sensing Technology for Military

Based on Fig. 4, it can be seen the process of obtaining the weight value of the Indicator of the effectiveness of remote sensing technology implementation in the military field. The

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process starts with identifying the strategic needs of the technology. By using base-evidence assessment, the assessment indicators are grouped based on Strategic Relevance, Operational Conditions, and Resource Limitations. The grouping of indicators can be seen in Table. I.

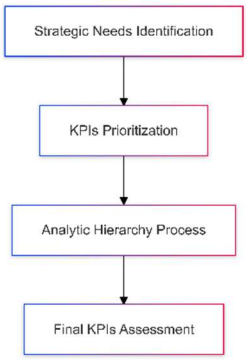

Fig. 4. Flow Diagram KPIs Assessment Process

TABLE I. INDICATOR GROUP

Indicator Group	Key Performance Indicators (KPIs)
Strategic Relevance	Accuracy
	Robustness and Reliability
	Scalability
Operational Conditions	Speed and Latency
	Flexibility and Adaptability
	Resource Utilization
Limited Resource	Energy Efficiency
	Cost-effectiveness

Furthermore, the indicator is calculated the weight value using the Analytic Hierarchy Process (AHP) method according to reference [9]. By giving AHP scale values to each indicator, the weight value is obtained as in Fig. 8. The main consideration of giving value to the AHP scale is the accuracy of the data generated by the system. The results of the AHP scale calculation consider the Consistency Ratio (CR) which is below 10%. With the CR obtained of 6.8%, the the matrix is considered consistent. From Fig. 5, global priorities are obtained which are used as weight values in the next calculation process. As for the determination of the score value of each indicator, five ranges were made from the tool specifications and base-evidence assessment from the literature study process. In general, these ranges are classified into a score range of 1-3 indicates low performance, 4-6 indicates moderate performance, 7-8 indicates high performance, and a score of 9-10 indicates optimal performance. Of course, this range can also be adjusted to the standards that will be applied in a particular environment. For the calculation of the weighted score, it is obtained using (1). The weighted scores are classified into three categories: a score of 8-10 signifies that the implementation is highly successful, a score of 6-7.9 signifies that the implementation is moderately successful with room for further development, and a score of less than 6 indicates that the implementation requires significant improvement.

$$Weighted\ Score = Score \times Weights \quad (1)$$

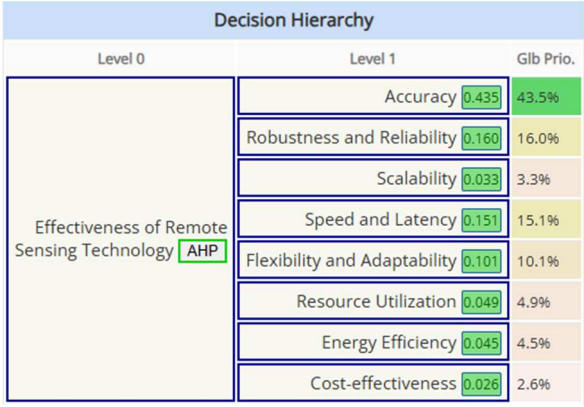

Fig. 5. Pairwise Comparison of All Criteria

III. RESULTS AND DISCUSSION

A. Type of Threats

Indonesia has a vast territory encompassing land and sea rich in natural resources. Such a large area directly implies the complexity of threats faced by the country, both military and non-military. The larger the area to be secured, the more diverse and complex the emerging threats become. These threats can originate from both internal and external sources and consist of both physical and non-physical threats. Physical threats, which are the primary focus of this discussion, include several important aspects such as terrorism, separatism, and border violations. These threats can come from both domestic and foreign sources and often have direct impacts on national security stability. Border violations represent a tangible threat that can occur at Indonesia's land or maritime borders. This becomes increasingly important as many border areas are located in hard-to-reach regions with limited surveillance capabilities, providing opportunities for those wishing to infiltrate or engage in illegal activities.

In terms of securing border areas, research has shown that maintaining sovereignty in these regions is not only crucial for national security but also for protecting economic and political interests. Effective surveillance at border regions can reduce potential economic losses caused by illegal activities such as smuggling or terrorism [10]. The vastness of Indonesia's territory encompassing land and sea gives rise to various physical and non-physical threats such as terrorism, separatism, and border violations. Therefore, effective surveillance at borders is essential for maintaining sovereignty as well as protecting national economic and political interests. In addition to physical threats, Indonesia also faces serious non-physical threats such as cyber-attacks from other countries. These cyber-attacks aim to weaken national defense systems, including critical infrastructure like communication networks and military information systems [11]. Given that digital technology has become the backbone of modern military operations, cyber threats can directly impact national security and cause significant losses if not properly addressed. The development and application of advanced technologies, especially remote sensing technology, are crucial in facing emerging threats whether they are physical or non-physical. Technologies like radar and satellite-based monitoring systems enable early detection and more effective surveillance in hard-to-access areas. According to [12], using remote sensing radar technology, especially Interferometric Synthetic Aperture Radar (InSAR), is very helpful for monitoring environmental changes at borders with high accuracy even

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under poor weather conditions or at night. This technology provides quicker detection capabilities and responses to emerging threats in border areas. The importance of remote sensing for real-time monitoring that can provide accurate information regarding potential military threats in hard-to-reach areas [13]. Information obtained from this technology is vital for strengthening national defense systems because it allows Indonesia's military to act more quickly and effectively against threats. Besides physical threats, Indonesia also faces serious cyber threats. Therefore, utilizing advanced technologies like remote sensing and radar is essential for early detection and real-time monitoring to strengthen national defense systems, especially in hard-to-reach areas. Moreover, [14] highlights the application of remote sensing technology in monitoring protected areas such as border regions prone to sovereignty violations and illegal activities. Remote sensing allows for detecting suspicious activities or environmental changes that may indicate a threat. In vast border areas often difficult for physical patrols to reach, this technology is crucial for enhancing surveillance effectiveness, especially in maintaining sovereignty and security in strategic regions. Adapting [14], Indonesia could implement remote sensing in border areas to maintain territorial integrity against external threats.

Furthermore, artificial intelligence (AI) plays an important role in supporting military operations on the ground. The development of AI-supported sensors that can assist military operators in decision-making on-site [15]. AI enables faster and more accurate data processing providing tactical advantages during critical situations which strengthens early detection capabilities as well as responses against both physical and non-physical threats. Additionally, AI technology is also utilized within cyber-security defense systems. Reference [16] shows that explainable AI becomes an essential tool for automating cyber-security measures and enhancing intelligence while building trust within digital defense systems. With increasing reliance on digital technologies, the use of AI within cyber-security can reduce vulnerabilities against attacks while strengthening Indonesia's defenses against cyber threats.

Given Indonesia's vast territory, the challenges faced are increasingly complex and varied. Both physical and non-physical threats require serious attention, especially with technological advancements available to strengthen national defenses. Remote sensing technology along with artificial intelligence are two key technologies that can help Indonesia tackle future security challenges, both in securing border areas as well as defending cyber-security systems. The development and integration of these technologies not only enhance border security but also contribute to national economic stability, as previously emphasized in earlier research [13], [17]. Remote sensing technology along with artificial intelligence (AI) plays a crucial role in strengthening Indonesia's defenses, particularly in monitoring borders while detecting both physical and cyber threats more quickly and accurately contributing towards national security as well as economic stability. Artificial Intelligence (AI) is increasingly becoming an important element in improving cyber defense mechanisms for military operations and digital security. One of the key aspects of AI applications is its capability in threat detection and response. AI and Machine Learning (ML) technologies, including Deep Learning (DL), are essential for analyzing large data sets to identify anomalies, predict cyber-attacks, and improve overall security posture [18], [19], [20]. Moreover, AI-enhanced cyber defense systems, such as the AI-Enhanced Cyber Defense System (AIECDS) methodology that includes data mapping and visualization, can simplify machine learning models and improve detection efficiency, even for threats with minimal data samples [21].

The effective scenarios for the application of AI in the military context include use in cyber operations to protect critical infrastructure and improve situational awareness [22], [23]. AI techniques such as reinforcement learning (RL) and deep RL are used to detect cyber-attacks and defend cyber-physical systems. However, while there are many benefits, the integration of AI in cyber-security also faces challenges such as potential AI failures, the need for constant updates, and lack of transparency [24], [25], [26]. Ethical considerations, including data privacy and bias, must also be addressed. The strategic implications of using AI in military operations include strategic decision-making and operational obfuscation, which can complicate defense missions while enhancing offensive capabilities [27], [28]. Thus, AI plays a significant role in strengthening cyber defense in military operations by improving threat detection, response, and proactive defense strategies [18], [22], [23], [29]. However, the challenges present require careful management to maximize the benefits of these technologies in the context of national security. To address the discussion arising from the use of AI in the military, [30] highlighted the importance of human control over AI to ensure responsibility, oversight, and accountability and advocated for a global framework for AI use that reflects international humanitarian law.

B. Remote Sensing Technology for Detecting Military Threats

The development of remote sensing technology for detecting threats in border areas has rapidly advanced with the integration of various advanced technologies. This surveillance system combines traditional technologies such as radar, Closed Circuit Television (CCTV), and infrared cameras with artificial intelligence (AI) and sensor networks to enhance the efficiency and accuracy of threat detection. The primary focus of this writing is to improve security in border areas through early detection of military threats, including infiltration by vehicles, drones, or unidentified objects. Radar is used to detect the presence of foreign objects approaching the border area. With radar, foreign objects attempting to approach can be detected from a relatively safe distance. As one of the main technologies, radar has undergone significant improvements with the integration of artificial intelligence. The use of AI enables radar not only to detect objects but also to identify and classify threats based on movement patterns. According to [31], AI-based radar allows for faster and more efficient detection of military objects such as drones and vehicles. AI also enhances the system's ability to predict threats based on historical pattern analysis, allowing for quicker responses in security operations. In the context of modern remote sensing, the use of technology in radar data processing has also become an important innovation, as described in [32], where AI assists in analyzing complex sensor data more efficiently. The development of remote sensing technology that integrates radar, CCTV, infrared cameras, artificial intelligence, and sensor networks has significantly improved early detection capabilities for military threats in border areas.

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CCTV continues to be a key component in surveillance at the borders, especially for monitoring hard-to-reach areas. The development of cloud computing-based CCTV allows for real-time surveillance and large-scale data processing. CCTV integrated with machine learning enables the system to detect threats with greater accuracy. According to [33], border surveillance technology continues to evolve, particularly with the utilization of cloud technology that allows for fast and efficient access and processing of data. Flying Ad Hoc Networks (FANETs), which combine CCTV sensors and drones, are very useful for monitoring areas that are physically difficult for ground troops to access, which is a major challenge in vast border regions like Indonesia [34]. Infrared cameras also play a vital role in military surveillance, especially at night or in adverse weather conditions. These cameras can detect body heat or vehicle engine heat that is not visible to regular cameras. Integrating data from various sensors, including radar, CCTV, and infrared cameras, provides greater power in effectively detecting threats [35]. The data generated by these infrared cameras can also be processed through AI algorithms to provide a clearer picture of threats that may not be detected by conventional systems. The use of cloud computing-based CCTV and machine learning, along with sensor integration such as infrared cameras and drones, has enhanced the effectiveness of threat detection in border areas, particularly in hard-to-reach locations.

To effectively apply radar, CCTV, and AI technologies, several practices and scenarios can be considered based on recent research and case studies. First, the integration of AI with radar can enhance radar systems through improved target detection, classification, and tracking [36], [37], [38]. AI-based radar systems can handle complex environments and interference better than traditional methods. In addition, the application of AI in CCTV systems enables the automation of infrastructure monitoring, thereby reducing the need for human intervention [39]. AI can track and analyze objects such as people and vehicles, making it ideal for high-security areas such as military bases and airports. Combining data from radar and CCTV can also improve surveillance accuracy, for example, the integration of radar with video data helps to better predict human trajectories [40]. One example of an effective scenario for the application of this technology includes port surveillance, where the integration of radar, CCTV, and AI can improve the detection of uncooperative vessels, thereby enhancing maritime security [41]. AI models trained on datasets such as VESSELimg can detect different types of vessels in real-time.

In addition to the technologies mentioned, the use of additional sensors such as WiFi radios, data loggers, and jammers further strengthens border surveillance systems. WiFi radios are used to enhance communication networks among sensors, allowing for rapid data transfer even in remote areas. According to [42], wireless network technology enables real-time communication between sensor devices and command centers, which is crucial in emergencies. Data loggers play a role in storing and recording surveillance data that can later be analyzed using stored historical data to predict future threats. Collecting and analyzing data through smart sensors provides deeper insights into enemy movement patterns, thus assisting the system in making threat predictions [31]. Signal jammers become an important technology for countering communication-based threats, especially from drones or enemy communication devices used for infiltration. Jammers can disrupt radio signals, GPS, or WiFi used by enemies, thereby hindering their operations. Adding jammers to remote sensing-based surveillance systems not only strengthens defenses but also disrupts enemy coordination along the border, giving security forces more time to respond [35]. Additionally, threats to cyber systems that rely on integrated remote sensing technology are also a significant concern. Reference [43] explains that anomaly detection strategies in cyber systems are essential for identifying threats to existing sensor infrastructure, particularly those based on networks. Utilizing artificial intelligence, sensor networks, and cloud computing enhances surveillance and response capabilities against threats. With the integration of various technologies, remote sensing not only aids in quickly and efficiently detecting threats but also strengthens defense systems with more accurate predictions and faster responses to infiltrations or enemy attacks in border areas.

C. Development of Defense Systems

The development of integrated and effective defense systems, utilizing various technologies such as remote sensing and the Internet of Things (IoT), is increasingly important in addressing complex security threats. Indonesia has developed the utilization of remote sensing technology for environmental monitoring and disaster management. Such as the collaboration between ESRI Indonesia and Indonesia's Army Topography Directorate (DITTOPAD) [44]. However, there is currently limited evidence explaining the application and use of real-time remote sensing technology for detecting military threats. The use of PIDS technology in Indonesia, which uses products from SENSTAR, is still limited to companies, embassies, and institutions [45], [46]. This gap highlights the need for further exploration and potential development of real-time remote sensing capabilities to address military concerns. Remote sensing technologies, such as satellites and optical sensors, play a crucial role in detecting physical movements and protecting territorial perimeters. Reference [1] highlights the importance of remote sensing in supporting international security through comprehensive and responsive surveillance capabilities against potential threats.

The development of defense systems using remote sensing in the context of border security, particularly Physical Intrusion Detection Systems (IDS) and Perimeter Intrusion Detection Systems (PIDS), is key to detecting and responding to physical threats. IDS are designed to detect threats approaching border areas through various sensor technologies, such as motion sensors, surveillance cameras, and other detection devices. Reference [47] emphasizes the importance of the rapid response offered by IDS to enhance border security. On the other hand, PIDS monitors the perimeter using optical sensors, radar, and other remote sensing technologies. The real-time monitoring capabilities of PIDS allow for quick preventive and responsive actions against potential threats [1]. The integration of remote sensing technologies, including satellites and optical sensors, along with IoT, enhances the effectiveness of these systems in detecting physical movements and protecting territorial perimeters. The development of integrated defense systems that leverage remote sensing technology and IoT is crucial for detecting and responding to complex security threats, especially in border security through the use of Physical Intrusion Detection Systems (IDS) and Perimeter Intrusion Detection Systems (PIDS).

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The use of IoT and information technology in IDS and PIDS systems also enables better data analysis and faster decision-making in critical situations. As explained in [48], the use of optical sensors and fiber optics has proven effective in detecting suspicious movements from a distance. The integration of these advanced technologies strengthens the effectiveness of border security systems, while [49] supports the effectiveness of optical sensors in enhancing the accuracy and detection range for threats. In border security strategies, the placement of defense systems is also a crucial aspect. The use of Geographic Information Systems (GIS) for analyzing the locations for deploying defense systems allows for identifying strategic points that consider range radius and potential entry points for threats [50]. This approach provides greater effectiveness in determining the locations for installing detection devices while considering confidentiality aspects to enhance surveillance. Additionally, advanced algorithms are increasingly being applied to optimize border surveillance. Reference [51] proposes using hybrid optimization algorithms that can improve the effectiveness of surveillance systems, while [14] demonstrates that remote sensing technology can provide important insights for identifying hidden threats. In this regard, reference [52] indicates that combining remote sensing with in-situ monitoring can identify complex spatial patterns. Understanding this is vital for designing more effective defense systems that are responsive to environmental changes that may affect detection capabilities and threat responses.

According to [52], integrating remote sensing with in-situ monitoring can provide more holistic information about environmental conditions at borders, including natural changes that may impact sensor effectiveness and monitoring devices. This is relevant in the context of Indonesia's border areas, where extensive geographical and environmental variations require adaptive and flexible surveillance approaches. For instance, in maritime border surveillance, remote sensing technology can identify suspicious sea current patterns that may indicate threats such as smuggling or other illegal activities. Furthermore, research in [53] shows that multi-sink wireless sensor networks can provide energy-efficient solutions for detecting intrusions at international borders and large military barracks. This highlights the importance of energy efficiency in the long-term operations of modern surveillance systems. Accuracy analysis methodologies in intelligent intrusion detection systems can enhance effectiveness and reliability [54], while [17] highlights the use of multi-level capsule networks for military object detection as a recent development in defense systems. On another note, [55] emphasizes the importance of filtering out irrelevant disturbances in fiber optic-based perimeter detection systems so that detections are more focused on actual threats. The integration of remote sensing with in-situ monitoring provides a more comprehensive understanding of border environmental conditions, which is essential for adaptive surveillance, while efficient sensor technologies and accuracy analysis methodologies enhance the effectiveness and reliability of intrusion detection systems against threats such as smuggling and illegal activities.

Reference [56] proposes using machine learning to enhance PIDS detection capabilities through well-curated datasets, while [57] suggests combining drones with static cameras for electromagnetic perimeter detection, providing higher security in monitored areas. With these various technological innovations, IDS and PIDS systems are increasingly capable of adapting to modern threats, both physical and cyber-related. Remote sensing technology and advanced algorithms play a vital role in supporting these efforts. As explained in [52], integrating remote sensing with direct monitoring provides a stronger foundation for understanding and responding to evolving threat dynamics across various geographic conditions. Therefore, investment in developing these technologies is crucial for maintaining security and sovereignty over the territory of the Unitary State of the Republic of Indonesia.

D. Objectives of Remote Sensing Technology Development

The development of remote sensing technology and advanced sensors has opened new horizons in the modern defense world, enabling significant improvements in the effectiveness of defense systems and enhancing the performance of existing technologies. In the context of defense, developing systems that utilize remote sensing technology is crucial for achieving several key aspects: deterrence, detection, defense, delay, and denial. Each of these aspects plays an important role in creating an effective defense system against military threats. Deterrence aims to prevent enemies from launching attacks by demonstrating the strength and capabilities of the defense system. Remote sensing technology plays a vital role in this by providing a clear picture of potential threats. As explained in [1], remote sensing can contribute to international security by supplying accurate data for threat analysis and strategic planning. The development of remote sensing technology within defense systems is essential because it enhances deterrence, detection, defense, delay, and denial. These capabilities contribute to the effectiveness of the defense system in preventing attacks, detecting threats early, protecting assets, slowing enemy advance, and denying unauthorized access.

Detection involves identifying threats early before they become dangerous. Research in [50] shows how the integration of remote sensing and GIS can improve ecosystem health detection, which can also be applied in monitoring military threats. Additionally, technologies such as optical and acoustic sensor systems can enhance intrusion detection over large areas [49], [58]. Reference [53] also proposes energy-efficient solutions for detecting intrusions at international borders using wireless sensor networks. Defense relates to the system's ability to protect areas or assets from detected threats. The development of technologies such as machine learning-based intrusion systems, as described in [56], enhances defense capabilities by using sensor data to detect and respond to threats in real-time. Research in [51] demonstrates that advanced algorithms can improve border surveillance effectiveness. Delay aims to slow down the advance of threats so that defensive responses can be implemented. Research in [54] with accuracy analysis methodologies shows how technology can slow intrusions and provide additional time for intervention. Reference [13] also discusses area monitoring technologies that can help delay potential threats by providing accurate information about enemy positions and movements. Deny involves the ability to eliminate threats or unwanted access. Advanced technologies, as outlined in [59], highlight the importance of denial strategies in maintaining military readiness. The development of defense systems that integrate sensor data and intelligent technologies is crucial to ensure that threats can be addressed effectively and efficiently [17], [31]. Early threat detection is a critical aspect of military defense that can be enhanced through the integration of remote sensing technology and GIS, as well as using optical and

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acoustic sensor systems for more effective threat detection, while machine learning-based systems strengthen responses to threats and maintain overall military readiness.

With the integration of remote sensing technology and advanced sensors, along with innovative data analysis methods, modern defense systems can better achieve their goals of deterrence, detection, defense, delay, and denial. Research and development in this field continue to evolve, with various studies exploring new ways to enhance defense system performance. Studies conducted in [6], [12], [32] outline how advancements in data analysis and remote sensing technology can improve the efficiency and effectiveness of defense systems. For example, the development of new algorithms utilizing machine learning and artificial intelligence (AI) in sensor data processing can significantly accelerate threat detection and identification. This is also discussed in [56], who highlight AI-based intrusion systems' capability to detect threats in real-time with high accuracy. Additionally, relevant supplementary references enrich our understanding of challenges and solutions in defense technology, including studies in [42] on cloud computing defense mechanisms and analyses in [60] on AI use in information warfare by major countries like the United States, China, and Russia. Thus, the development and application of remote sensing technology in a defense context is not just about technical innovation but also about building integrated and adaptive strategies to face increasingly dynamic and complex military threats. Continuous research and development in this field are essential to ensure that defense systems can continually adapt to technological advancements and evolving threat characteristics. An integrated strategy will be key to creating a more resilient, responsive, and effective defense system for national security protection.

E. Challenges in Developing Remote Sensing-Based Defense Systems

In the implementation phase of defense technology development, one of the primary challenges is the supporting infrastructure required. Advanced technologies necessitate a stable network for real-time monitoring, which aligns with the findings in [61] that real-time data management requires reliable infrastructure for the system to function optimally. Additionally, significant investments are needed in the procurement of hardware and software, as identified in [62] regarding the importance of surveillance in large-scale infrastructure investments in support of long-term sustainability. Technologies in border areas face even greater challenges related to infrastructure and security. Reference [33] explains that border control technologies play a crucial role in securing the region from external threats. To overcome infrastructure challenges and implement military personnel training programs with remote sensing technologies, several strategies can be adopted. The first strategy is to integrate remote sensing technology with existing infrastructure to improve monitoring and management [63]. For example, satellite remote sensing can be used in grid line and substation planning, overhead transmission line inspection, and disaster emergency monitoring in the electricity system. In addition, remote sensing offers high efficiency and low cost, thereby reducing the need for on-site personnel presence, which is critical in maintaining infrastructure in challenging environments [63]. The synergy between the use of this technology for military reconnaissance and civilian infrastructure projects also shows the potential for beneficial dual use [64].

Apart from the technical aspects, leadership challenges are also a significant concern. As explained in [76], organizational leaders must be able to address the challenges of implementing new technologies, both in terms of internal and external coordination. Therefore, a phased approach is necessary. Technological development cannot be done hastily but requires a gradual and sustainable process to maximize available resources. Reference [77] emphasizes the importance of investing in technological innovation to address natural resource challenges, indicating that a step-by-step and sustainable approach is key to success. Strategic leadership plays a critical role in the successful implementation of technology. Reference [78] reveals that effective leadership is a critical factor in technological innovation because leaders can promote the adoption of new technologies through strategic decisions and political support. Similarly, [79] shows that strategic investment decisions in new technologies must be based on deep analysis, particularly in sectors that require technology-based approaches such as health, and can be applied to defense and security. Effective leadership is also necessary in addressing the challenges of implementing technology in border areas. Reference [80] stresses that digital transformation presents challenges and opportunities for leaders, especially in developing countries that must respond to new dynamics, including those related to pandemics. In this context, leaders in Indonesia's border regions need to consider how new technologies can support security while ensuring that all necessary elements, such as training and personnel readiness, are met. Advanced technologies in Indonesia's border regions not only require the right infrastructure but also trained and ready human resources. Reference [81] highlights the importance of human resource information systems in military environments to support operational effectiveness. Without adequate training, field personnel may not be able to operate these technologies efficiently. Therefore, regular training and the provision of appropriate training resources are crucial, as proposed in [82], who suggest that investments in technological infrastructure must be accompanied by the development of workforce skills to optimize innovation.

In terms of military personnel training programs, it is important to develop a comprehensive training plan that covers various remote sensing applications, such as satellite remote sensing, terrain remote sensing, as well as the integrated techniques of Remote Sensing (RS), Global Positioning System (GPS), and Geographic Information System (GIS) [83]. This approach ensures that personnel have a deep understanding of the various aspects of these technologies. In addition, integrating practical exercises and real-world applications into training programs can significantly improve learning outcomes, an example of which is NASA's ARSET program which showed that practice-based training improved knowledge and decision-making capabilities [84]. Training should also emphasize the use of remote sensing data in the decision-making process by simulating real-world scenarios where the data are used for operational command and battlefield management [85]. For the implementation of this strategy, leveraging existing training programs such as NASA's Applied Remote Sensing Training Program (ARSET) can be an effective first step [84]. In another case, collaboration with virtual centers of excellence will provide access to cutting-edge research and expertise in remote sensing technologies [86]. Finally, it is

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important to regularly update training content to cover the latest advances in these technologies and be able to address new challenges in military operations [86]. This study recommends dividing the development of human resources into six milestones to ensure that each stage of development is measurable. Each milestone has specific objectives and goals that are interconnected, and aimed at preparing personnel to effectively utilize the technology that has been developed. For further clarity, please refer to the following Table II.

TABLE II. HUMAN RESOURCE DEVELOPMENT MILESTONES

Milestone	Timeline	Output
Analysis & Program Preparation	Milestone 1	Training curriculum, needs analysis, budget readiness.
Basic Training & Cyber-defense	Milestone 2	Basic understanding of technology, initial evaluation.
Advanced Training & Soft Skills	Milestone 3	Advanced competency, real-case scenario simulations.
Certification & Final Assessment	Milestone 4	Certificates and technology operation licenses.
Formation of Technology Driving Team	Milestone 5	Formation of the technology-driving team.
Evaluation & Program Refinement	Milestone 6	Evaluation report and training plan for the next year.

In addition to technical and leadership aspects, political support and strategic decisions from leaders are essential to ensure that technological investments are not only made but also implemented effectively. Strategic and innovation-focused leadership is crucial for driving change within organizations [78]. In practice, the involvement of stakeholders such as military personnel, government officials, and technology providers is crucial in the technological development process to provide insight and promote collaboration focused on the successful implementation of remote sensing technology for defense purposes. Therefore, this research suggests assigning roles to the three primary stakeholders in Table III, thereby facilitating their coordinated efforts toward achieving effective integration of remote sensing technology into defense strategies.

TABLE III. ASSIGNING ROLES

Activity	Role
Threat Identification	Military; Government
SOP Development	Military; Government
Technology Developing and Training	Technology Provider
Response Simulation	Military
Technology Installation	Technology Provider
SOP Integration	Military; Technology Provider
Real-time Monitoring	Military
Data Analysis	Military
Execute Response Procedures	Military
Situation Coordination	Government
Procedure Review	Military; Government; Technology Provider
Continuous Improvement	Military; Government; Technology Provider

Overall, the implementation of advanced technologies to support defense systems in Indonesia's border regions requires a comprehensive and adaptive approach to the specific challenges faced in the field. Apart from infrastructure investments, strategic leadership is another critical factor. Leaders with long-term vision and a deep understanding of technology play a key role in the successful implementation of defense technologies. With strong leadership support and gradual investments, the challenges can be overcome, and the effectiveness of the defense system can be enhanced. Additional challenges arise in terms of the policy support needed to ensure that technological developments align with broader national strategies. Clear regulations and consistent political support will be crucial in ensuring the long-term success of remote sensing-based defense technologies, as without clear direction from policymakers, various technological initiatives often stall. Moreover, trained and ready human resources are a determining factor in the success of implementation in the defense sector. Another suggestion is the collaboration between public and private sectors in addressing the challenges of defense technology development. The role of the private sector in providing technology, innovation, and creative solutions for defense systems can act as a catalyst in accelerating the adoption of advanced technologies. This collaboration will also help reduce investment costs, as the private sector can share the burden with the government in terms of research, development, and implementation of technologies. Involving various stakeholders in the investment and development process will strengthen synergy and knowledge, which in turn can enhance the effectiveness of technology implementation. With strong leadership support, gradual investments, and good collaboration, all these factors are crucial in addressing the challenges faced and ultimately ensuring that the adopted technologies provide optimal benefits for national defense.

F. Implementing Remote Sensing-Based Defense Systems

To effectively evaluate the implementation of real-time remote sensing technology, several key performance indicators (KPIs) should be considered. First, accuracy is a crucial factor, encompassing the precision of the data collected and processed, particularly in terms of object detection and classification in remote sensing images [65], [66], [67]. Next, speed and latency are also of paramount importance. The time required to process and analyze data in real-time can significantly impact the system's effectiveness, including the speed of data acquisition and the latency in delivering actionable insights [67], [68], [69]. Additionally, the energy efficiency of the hardware used for real-time processing is an important metric, especially in resource-limited environments such as Unmanned Aerial Vehicles (UAVs) and edge devices [69], [70]. The scalability of the system must also be considered. The ability to handle increasing amounts of data and to scale operations without significant loss of performance is key to long-term success [71], [72]. Furthermore, the robustness and reliability of the system under varying conditions, such as different weather scenarios, as well as its ability to maintain consistent output, are critical considerations [67]. Resource utilization, including efficient use of CPU, GPU, and memory, is essential for optimizing system performance [70], [71]. Moreover, the flexibility and adaptability of the system to adjust to different remote sensing tasks and environments, such as various types of sensors and platforms, are important aspects that should not be overlooked [69], [73]. Finally, the cost-effectiveness of

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implementing and maintaining the real-time remote sensing system must be thoroughly evaluated, encompassing hardware, software, and operational costs [74]. By considering all these KPIs, we can establish a comprehensive framework for assessing the effectiveness and efficiency of real-time remote sensing technology implementations, ensuring that these systems meet required standards and operational needs. Based on these KPIs, an evaluation matrix can be created as shown in Table IV. The table includes KPI parameters, weights, scores, and weighted scores.

By obtaining parameters to evaluate the feasibility of implementing a real-time defense system, it can be simulated with a case study on border guarding between Indonesia and Malaysia in Betung Kerihun National Park, West Kalimantan [75]. Some of the data in this study comes from sources that cannot be disclosed in order to maintain confidentiality. To avoid conflicts in the future, simulated data is used that is based on the data. The conditions determined to carry out the simulation include the length of the border ± 368 km, internet and electricity infrastructure conditions are in poor condition, the topographic location of the border is hilly and dense vegetation, weather conditions are often rainy, and there are 478 State Border Pillars.

TABLE IV. KEY PERFORMANCE INDICATORS (KPIS) SCORING

Key Performance Indicators (KPIs)	Weights (%)	Score (1-10)	Weighted Score (%)
Strategic Indicators			
Accuracy	43.5	6	2.61
Robustness and Reliability	16.0	5	0.80
Scalability	3.3	4	0.13
Operational Conditions			
Speed and Latency	15.1	5	0.76
Flexibility and Adaptability	10.1	6	0.61
Resource Utilization	4.9	5	0.25
Limited Resources			
Energy Efficiency	4.5	4	0.18
Cost-effectiveness	2.6	5	0.13
Total	100	-	5.17

Based on the results of the total weight calculation, with a value of 5.17%, indicates that the implementation requires significant improvement. From Table V it can be seen that the cost of deploying troops to guard the border between Indonesia and Malaysia in Betung Kerihun National Park is Rp. 5.04 billion over five years, which is cheaper than any intrusion system technology. Despite this low cost, human surveillance has significant limitations. One of the main challenges is the effectiveness of coverage, where personnel find it difficult to monitor the entire 368 km wide area. In addition, fatigue and consistency are issues, as human performance can degrade in difficult conditions such as bad weather and hilly terrain. Although more expensive, these technology offers higher effectiveness in real-time detection and automation.

TABLE V. COST COMPARISON

System	Installation Cost (Rp)	5 Years Operation Cost (Rp)	5 Years Total Cost (Rp)
Advanced System	8.28 trillion	2.07 trillion	10.35 trillion
Drone & IoT System	828 billion	414 billion	1.24 trillion
Low Cost System	414 billion	310.5 billion	724.5 billion
Troops Deployment	-	-	5.04 billion

IV. CONCLUSIONS

The importance of the creation of a reliable remote sensing system for the immediate identification of military threats in Indonesia is the focus of this study. The study demonstrates how the incorporation of cutting-edge technologies such as artificial intelligence (AI), radar, and CCTV can significantly improve the ability to detect both physical and non-physical threats, including cyber-attacks, that pose a threat to national security. The study finds that while remote sensing technologies offer promising approaches to border threat detection, barriers such as poor infrastructure and personnel training prevent their widespread use. Addressing these issues is essential to maximizing the performance of defense systems. Strategic investments in communications infrastructure, such as Base Transceiver Stations (BTS) and programs like PALAPA RING, are needed to improve surveillance capabilities in hard-to-reach border areas. On the basis of simulation data, it has been found that the implementation of remote sensing technology for border areas can be cost-effective.

The study also highlights how effective border monitoring protects economic interests by reducing illegal activities such as terrorism and smuggling, as well as military threats. The findings support a holistic strategy that strengthens physical surveillance and cyber-security measures to provide a multi-layered defense. In summary, Indonesia can significantly increase its readiness to respond to multiple threats by overcoming infrastructure barriers and improving personnel training. Through the integration of advanced remote sensing technologies, this research provides important insights into how national defense strategies can be further developed, ultimately leading to an increase in the security and stability of the country.

As for the weighting of KPIs can be explored further, for example by considering other indicators such as Cost-effectiveness or Scalability, given that Indonesia's vast territory requires the most up-to-date testing models before implementation. To overcome the challenges that arise such as infrastructure, human resources and large financing, further studies on suitable strategies are very useful. For example, ensuring that public-private partnerships are established to optimize resources, continuous training for personnel can be enhanced, or implementation strategies for pilot projects. Additional studies on remote sensing systems for military threat detection should focus on several key areas to improve their effectiveness. First, there is a need to explore advanced sensor technologies, particularly the integration of multi-sensor systems that combine radar, infrared, and optical devices to improve detection accuracy in diverse environments. In addition, cybersecurity remains a critical concern. Therefore, the development of robust frameworks to

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protect remote sensing technologies from cyber threats is essential. Research should also address infrastructure development, exploring cost-effective deployment models suitable for Indonesia's challenging terrain and ensuring that public-private partnerships are established to optimize resources. It is also important to develop comprehensive national defense policies that incorporate advances in remote sensing, while exploring opportunities for international cooperation in sharing technology.

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REFERENCES

[1] R. Avtar *et al.*, "Remote sensing for international peace and security: Its role and implications," *Remote Sens.*, vol. 13, no. 3, 2021, doi: 10.3390/rs13030439.

[2] S. Wang, W. Xu, and T. Guo, "Advances in Thermal Infrared Remote Sensing Technology for Geothermal Resource Detection," *Remote Sens.*, vol. 16, no. 10, 2024, doi: 10.3390/rs16101690.

[3] D. M. Hargitai and A. Bencsik, "The Role of Leadership in Digital Learning Organizations," *Emerg. Sci. J.*, vol. 7, no. Special issue 2, pp. 111–124, 2023, doi: 10.28991/ESJ-2023-SIED2-09.

[4] A. Tricot and J. Sweller, "Domain-Specific Knowledge and Why Teaching Generic Skills Does Not Work," 2014. doi: 10.1007/s10648-013-9243-1.

[5] J. Dalzochio *et al.*, "Predictive Maintenance in the Military Domain: A Systematic Review of the Literature," *ACM Comput. Surv.*, vol. 55, no. 13 s, 2023, doi: 10.1145/3586100.

[6] Y. J. Park and S. M. Jones-Jang, "Surveillance, security, and AI as technological acceptance," *AI Soc.*, vol. 38, no. 6, 2023, doi: 10.1007/s00146-021-01331-9.

[7] BAKTI, "Layanan." Accessed: Nov. 18, 2024. [Online]. Available: https://baktikominfo.id/layanan

[8] DJPPI, "DJPPI Kominfo Merilis Infografis Infrastruktur Jaringan Telekomunikasi Indonesia." Accessed: Nov. 18, 2024. [Online]. Available: https://djppi.kominfo.go.id/news/djppi-kominfo-merilis-infografis-infrastruktur-jaringan-telekomunikasi-indonesia

[9] K. Goepel, "Implementation of an Online software tool for the Analytic Hierarchy Process (AHP-OS)," *Int. J. Anal. Hierarchy Process*, vol. 10, no. 3, pp. 469–487, 2018, doi: 10.13033/ijahp.v10i3.590.

[10] N. Lori and K. Schilde, "A Political Economy of Global Security Approach to Migration and Border Control," 2021. doi: 10.1093/jogss/ogaa011.

[11] Y. Li and Q. Liu, "A comprehensive review study of cyber-attacks and cyber security; Emerging trends and recent developments," *Energy Reports*, vol. 7, pp. 8176–8186, Nov. 2021, doi: 10.1016/J.EGYR.2021.08.126.

[12] M. Motagh and B. Riedel, "Editorial for Special Issue: RADAR REMOTE SENSING," *PFG - J. Photogramm. Remote Sens. Geoinf. Sci.*, vol. 92, no. 3, p. 211, 2024, doi: 10.1007/s41064-024-00293-z.

[13] B. Zeng, S. Gao, Y. Xu, Z. Zhang, F. Li, and C. Wang, "Detection of Military Targets on Ground and Sea by UAVs with Low-Altitude Oblique Perspective," *Remote Sens.*, vol. 16, no. 7, pp. 1–24, 2024, doi: 10.3390/rs16071288.

[14] Y. Wang, Z. Lu, Y. Sheng, and Y. Zhou, "Remote Sensing Applications in Monitoring of Protected Areas," *Remote Sens.*, vol. 12, no. 9, 2020, doi: 10.3390/rs12091370.

[15] B. K. Russell, J. McGeown, and B. L. Beard, "Developing AI enabled sensors and decision support for military operators in the field," 2023. doi: 10.1016/j.jsams.2023.03.001.

[16] I. H. Sarker, H. Janicke, A. Mohsin, A. Gill, and L. Maglaras, "Explainable AI for cybersecurity automation, intelligence and trustworthiness in digital twin: Methods, taxonomy, challenges and prospects," *ICT Express*, vol. 10, no. 4, pp. 935–958, Aug. 2024, doi: 10.1016/J.ICTE.2024.05.007.

[17] B. Janakiramaiah, G. Kalyani, A. Karuna, L. V. N. Prasad, and M. Krishna, "Military object detection in defense using multi-level capsule networks," *Soft Comput.*, vol. 27, no. 2, 2023, doi: 10.1007/s00500-021-05912-0.

[18] F. Amato *et al.*, "AI for Sustainability: Activities of the CINI-AIIS Lab at University of Naples Federico II," *CEUR Workshop Proc.*, vol. 3762, pp. 522–527, 2024.

[19] M. Macas and C. Wu, "Review: Deep Learning Methods for Cybersecurity and Intrusion Detection Systems," *Proc. - 2020 IEEE Latin-American Conf. Commun. LATINCOM 2020*, p. 9282324, 2020, doi: 10.1109/LATINCOM50620.2020.9282324.

[20] D. S. Bogatinov, M. Bogdanoski, and S. Angelevski, "AI-based cyber defense for more secure cyberspace," *Nature-Inspired Comput. Concepts, Methodol. Tools, Appl.*, vol. 3–3, pp. 1471–1489, 2016, doi: 10.4018/978-1-5225-0788-8.ch056.

[21] C. Klopper and J. H. P. Eloff, "Data fingerprinting and visualization for AI-Enhanced Cyber-Defence Systems," *IEEE Access*, p. 3482728, 2024, doi: 10.1109/ACCESS.2024.3482728.

[22] N. E. Fard, R. R. Selmic, and K. Khorasani, "A Review of Techniques and Policies on Cybersecurity Using Artificial Intelligence and Reinforcement Learning Algorithms," *IEEE Technol. Soc. Mag.*, vol. 42, no. 3, pp. 57–68, 2023, doi: 10.1109/MTS.2023.3306540.

[23] K. W. Bennett and J. Robertson, "An AI-based framework for remote sensing supporting multi-domain operations," p. 16, 2022, doi: 10.1117/12.2618923.

[24] U. U. Ibekwe, U. M. Mbanaso, and N. A. Nnanna, "A Critical Review of The Intersection of Artificial Intelligence and Cybersecurity," *2023 2nd Int. Conf. Multidiscip. Eng. Appl. Sci. ICMEAS 2023*, p. 10379362, 2023, doi: 10.1109/ICMEAS58693.2023.10379362.

[25] S. M. Nour and S. A. Said, "Harnessing the Power of AI for Effective Cybersecurity Defense," *6th Int. Conf. Comput. Informatics, ICCI 2024*, pp. 98–102, 2024, doi: 10.1109/ICCI61671.2024.10485059.

[26] R. Montasari, "Internet of Things and Artificial Intelligence in National Security: Applications and Issues," *Adv. Inf. Secur.*, vol. 101, pp. 27–56, 2023, doi: 10.1007/978-3-031-21920-7_3.

[27] C. Whyte, "Problems of Poison: New Paradigms and 'agreed' Competition in the Era of AI-Enabled Cyber Operations," *Int. Conf. Cyber Conflict, CYCON*, vol. 2020-May, pp. 215–232, 2020, doi: 10.23919/CyCon49761.2020.9131717.

[28] J. Burton and S. R. Soare, "Understanding the Strategic Implications of the Weaponization of Artificial Intelligence," *Int. Conf. Cyber Conflict, CYCON*, vol. 2019-May, p. 8756866, 2019, doi: 10.23919/CYCON.2019.8756866.

[29] Simran, S. Kumar, and A. Hans, "The AI Shield and Red AI

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Framework: Machine Learning Solutions for Cyber Threat Intelligence(CTI),” *2024 Int. Conf. Intell. Syst. Cybersecurity, ISCS 2024*, p. 10581195, 2024, doi: 10.1109/ISCS61804.2024.10581195.

[30] F. L. López and A. P. Gutiérrez, “Framework of Public International Law and Military Uses of Artificial Intelligence in the Eu,” *Rev. Electron. Estud. Int.*, no. 46, pp. 505–526, 2023, doi: 10.36151/reei.46.17.

[31] A. Bin Rashid, A. K. Kausik, A. Al Hassan Sunny, and M. H. Bappy, “Artificial Intelligence in the Military: An Overview of the Capabilities, Applications, and Challenges,” *Int. J. Intell. Syst.*, vol. 2023, 2023, doi: 10.1155/2023/8676366.

[32] A. A. Aleissaee *et al.*, “Transformers in Remote Sensing: A Survey,” *Remote Sens.*, vol. 15, no. 7, pp. 1–31, 2023, doi: 10.3390/rs15071860.

[33] N. Amelung and V. Galis, “Border control technologies: introduction,” *Sci. Cult. (Lond).*, vol. 32, no. 3, pp. 323–343, 2023, doi: 10.1080/09505431.2023.2234932.

[34] F. Al Fayez, M. Hammoudeh, B. Adebisi, and K. N. Abdul Sattar, “Assessing the effectiveness of flying ad hoc networks for international border surveillance,” *Int. J. Distrib. Sens. Networks*, vol. 15, no. 7, Jul. 2019, doi: 10.1177/1550147719860406/ASSET/IMAGES/LARGE/10.1177_1550147719860406-FIG2.JPEG.

[35] L. Patino *et al.*, “Fusion of Heterogenous Sensor Data in Border Surveillance,” *Sensors*, vol. 22, no. 19, pp. 1–17, 2022, doi: 10.3390/s22197351.

[36] M. K. Ha *et al.*, “Comparative Analysis of Audio Processing Techniques on Doppler Radar Signature of Human Walking Motion Using CNN Models,” *Sensors (Basel).*, vol. 23, no. 21, p. 23218743, 2023, doi: 10.3390/s23218743.

[37] A. Wrabel, R. Graef, and T. Brosch, “A Survey of Artificial Intelligence Approaches for Target Surveillance with Radar Sensors,” *IEEE Aerosp. Electron. Syst. Mag.*, vol. 36, no. 7, pp. 26–43, 2021, doi: 10.1109/MAES.2021.3065069.

[38] H. Deng and B. Himed, “Target Detection and Interference Mitigation in Future AI-Based Radar Systems,” *IEEE Natl. Radar Conf. - Proc.*, vol. 2021-May, p. 9455198, 2021, doi: 10.1109/RadarConf2147009.2021.9455198.

[39] F. Alam and A. Alraeesi, “Computer Vision Based Smart CCTV Solution,” *Soc. Pet. Eng. - ADIPEC 2022*, p. 211842, 2022, doi: 10.2118/211842-MS.

[40] T. Fernando, S. Denman, S. Sridharan, and C. Fookes, “Pedestrian Trajectory Prediction with Structured Memory Hierarchies,” in *Machine Learning and Knowledge Discovery in Databases*, M. Berlingerio, F. Bonchi, T. Gärtner, N. Hurley, and G. Ifrim, Eds., Cham: Springer International Publishing, 2019, pp. 241–256.

[41] B. Rubí *et al.*, “VESSELimg: A Large UAV-Based Vessel Image Dataset for Port Surveillance,” *2024 Int. Conf. Unmanned Aircr. Syst. ICUAS 2024*, pp. 76–83, 2024, doi: 10.1109/ICUAS60882.2024.10556944.

[42] S. El Kafhali, I. El Mir, and M. Hanini, “Security Threats, Defense Mechanisms, Challenges, and Future Directions in Cloud Computing,” *Arch. Comput. Methods Eng.*, vol. 29, 2022, doi: 10.1007/s11831-021-09573-y.

[43] N. Jeffrey, Q. Tan, and J. R. Villar, “A Review of Anomaly Detection Strategies to Detect Threats to Cyber-Physical Systems,” *Electron.*, vol. 12, no. 15, pp. 1–34, 2023, doi: 10.3390/electronics12153283.

[44] ESRI Indonesia, “Dittopad 4.0: Supporting Indonesia’s national security operations.” Accessed: Dec. 05, 2024. [Online]. Available: https://esriindonesia.co.id/news/dittopad-40-supporting-indonesias-national-security-operations

[45] Aetherica, “Our Clients.” Accessed: Dec. 05, 2024. [Online]. Available: https://www.aetherica.com/our-clients.html

[46] SENSTAR, “What is Perimeter Intrusion Detection System (PIDS)?” [Online]. Available: https://senstar.com/senstarpedia/what-is-perimeter-intrusion-detection-system-pids/

[47] Z. A. Khan and P. Herrmann, “Recent advancements in intrusion detection systems for the internet of things,” *Secur. Commun. Networks*, vol. 2019, 2019, doi: 10.1155/2019/4301409.

[48] N. P. Goodman, “Border militarization and domestic institutions,” *Const. Polit. Econ.*, no. 0123456789, 2024, doi: 10.1007/s10602-024-09440-5.

[49] A. Vikram, S. K. Patel, A. Chaturvedi, O. Alsalman, and J. Parmar, “Detecting accurate parametric intrusions using optical fiber sensors for long-distance data communication system,” *Opt. Fiber Technol.*, vol. 80, p. 103453, Oct. 2023, doi: 10.1016/J.YOFTE.2023.103453.

[50] I. Soubry, T. Doan, T. Chu, and X. Guo, “A systematic review on the integration of remote sensing and gis to forest and grassland ecosystem health attributes, indicators, and measures,” *Remote Sens.*, vol. 13, no. 16, 2021, doi: 10.3390/rs13163262.

[51] M. Tariq, A. Saadat, R. Ahmad, Z. Abaid, and J. J. P. C. Rodrigues, “Enhanced Border Surveillance Through a Hybrid Swarm Optimization Algorithm,” *IEEE Sens. J.*, vol. 23, no. 22, 2023, doi: 10.1109/JSEN.2023.3317531.

[52] F. Weiser *et al.*, “Combining in-situ monitoring and remote sensing to detect spatial patterns of volcanic sulphur impact on pine needles,” *For. Ecol. Manage.*, vol. 549, p. 121468, Dec. 2023, doi: 10.1016/J.FORECO.2023.121468.

[53] K. Ghosh, S. Neogy, P. K. Das, and M. Mehta, “Intrusion Detection at International Borders and Large Military Barracks with Multi-sink Wireless Sensor Networks: An Energy Efficient Solution,” *Wirel. Pers. Commun.*, vol. 98, no. 1, 2018, doi: 10.1007/s11277-017-4909-5.

[54] X. Hu, T. Li, Z. Wu, X. Gao, and Z. Wang, “Research and application of intelligent intrusion detection system with accuracy analysis methodology,” *Infrared Phys. Technol.*, vol. 88, pp. 245–253, Jan. 2018, doi: 10.1016/J.INFRARED.2017.11.032.

[55] S. S. Mahmoud, “Practical Aspects of Perimeter Intrusion Detection and Nuisance Suppression for Distributed Fiber-Optic Sensors,” *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, doi: 10.1109/TIM.2023.3284133.

[56] S. Pitafi, T. Anwar, I. D. M. Widia, B. Yimwadsana, and S. Pitafi, “Revolutionizing Perimeter Intrusion Detection: A Machine Learning-Driven Approach with Curated Dataset Generation for Enhanced Security,” *IEEE Access*, vol. 11, 2023, doi: 10.1109/ACCESS.2023.3318600.

[57] P. Teixidó *et al.*, “Secured perimeter with electromagnetic detection and tracking with drone embedded and static cameras,” *Sensors*, vol. 21, no. 21, 2021, doi: 10.3390/s21217379.

[58] N. Al-Khalli, S. Alateeq, M. Almansour, Y. Alhassoun, A. B. Ibrahim, and S. A. Alshebeili, “Real-Time Detection of Intruders Using an Acoustic Sensor and Internet-of-Things Computing,” *Sensors*, vol. 23, no. 13, pp. 1–16, 2023, doi: 10.3390/s23135792.

[59] I. Peets, “The hidden cost of excess capabilities: a threat to future

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military readiness," *Def. Secur. Anal.*, vol. 40, no. 2, pp. 211–234, 2024, doi: 10.1080/14751798.2024.2325220.

[60] L. Y. Hunter, C. D. Albert, J. Rutland, K. Topping, and C. Hennigan, "Artificial intelligence and information warfare in major power states: how the US, China, and Russia are using artificial intelligence in their information warfare and influence operations," *Def. Secur. Anal.*, vol. 40, no. 2, 2024, doi: 10.1080/14751798.2024.2321736.

[61] T. P. da Costa, D. M. B. da Costa, and F. Murphy, "A systematic review of real-time data monitoring and its potential application to support dynamic life cycle inventories," *Environ. Impact Assess. Rev.*, vol. 105, p. 107416, Mar. 2024, doi: 10.1016/J.EIAR.2024.107416.

[62] Y. Wu *et al.*, "Investment monitoring key points identification model of big science research infrastructures -- Fuzzy BWM-entropy-PROMETHEE II method," *Socioecon. Plann. Sci.*, vol. 86, p. 101461, Apr. 2023, doi: 10.1016/J.SEPS.2022.101461.

[63] W. Tong *et al.*, "Application Research of Satellite Remote Sensing Technology in Power System," *2019 5th Int. Conf. Control. Autom. Robot. ICCAR 2019*, pp. 6–10, 2019, doi: 10.1109/ICCAR.2019.8813736.

[64] Z. Wang, Q. Kang, Y. Xun, Z. Shen, and C. Cui, "Military reconnaissance application of high-resolution optical satellite remote sensing," *Int. Symp. Optoelectron. Technol. Appl. 2014 Opt. Remote Sens. Technol. Appl.*, vol. 9299, p. 929919, 2014, doi: 10.1117/12.2073125.

[65] Y. Deng, F. Bi, L. Chen, and T. Long, "Research on high-performance remote sensing image real-time processing system," *2010 Int. Conf. Comput. Des. Appl. ICCDA 2010*, vol. 1, p. 5540848, 2010, doi: 10.1109/ICCDA.2010.5540848.

[66] H. Hashmi, R. Dwivedi, and A. Kumar, "YOLO-RS: An Efficient YOLO-Based Approach for Remote Sensing Object Detection," *Proc. 2023 12th Int. Conf. Syst. Model. Adv. Res. Trends, SMART 2023*, pp. 50–56, 2023, doi: 10.1109/SMART59791.2023.10428504.

[67] L. Bian, J. Wang, B. Li, and Y. Shi, "Improved real time object detection method for remote sensing image based on YOLOv4," p. 167, 2022, doi: 10.1117/12.2661016.

[68] H. Ding *et al.*, "A Novel Mamba Architecture with a Semantic Transformer for Efficient Real-Time Remote Sensing Semantic Segmentation," *Remote Sens.*, vol. 16, no. 14, p. 16142620, 2024, doi: 10.3390/rs16142620.

[69] Z. Cao, L. Kooistra, W. Wang, L. Guo, and J. Valente, "Real-Time Object Detection Based on UAV Remote Sensing: A Systematic Literature Review," *Drones*, vol. 7, no. 10, p. 7100620, 2023, doi: 10.3390/drones7100620.

[70] X. Wei, W. Liu, L. Chen, L. Ma, H. Chen, and Y. Zhuang, "FPGA-based hybrid-type implementation of quantized neural networks for remote sensing applications," *Sensors (Switzerland)*, vol. 19, no. 4, p. 19040924, 2019, doi: 10.3390/s19040924.

[71] M. K. Pekturk and M. Unal, "A review on real-time big data analysis in remote sensing applications," *2017 25th Signal Process. Commun. Appl. Conf. SIU 2017*, p. 7960499, 2017, doi: 10.1109/SIU.2017.7960499.

[72] A. Plaza, Q. Du, Y. L. Chang, and R. L. King, "High Performance Computing for Hyperspectral Remote Sensing," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 4, no. 3, pp. 528–544, 2011, doi: 10.1109/JSTARS.2010.2095495.

[73] C. Li *et al.*, "A comprehensive calibration site for high resolution remote sensors dedicated to quantitative remote sensing and its applications," *Natl. Remote Sens. Bull.*, vol. 25, no. 1, pp. 198–219, 2021, doi: 10.11834/jrs.20210326.

[74] C. F. Schueler, "Satellite-instrument system engineering best practices and lessons," *Remote Sens. Syst. Eng. II*, vol. 7458, p. 745807, 2009, doi: 10.1117/12.829262.

[75] KSDAE, "Investigasi Pilar Batas Negara RI-Malaysia di Wilayah TN Betung Kerihun." Accessed: Dec. 06, 2024. [Online]. Available: https://ksdae.menlhk.go.id/info/6254/investigasi-pilar-batas-negara-ri-malaysia-di-wilayah-tn-betung-kerihun.html

[76] L. Haber and A. Carmeli, "Leading the challenges of implementing new technologies in organizations," *Technol. Soc.*, vol. 74, p. 102300, Aug. 2023, doi: 10.1016/J.TECHSOC.2023.102300.

[77] M. Murshed, "Exploring the relevance of investing in technological innovation programs for tackling natural resource consumption-related environmental challenges in developing countries," *Environ. Challenges*, vol. 14, p. 100844, Jan. 2024, doi: 10.1016/J.ENVC.2024.100844.

[78] C. Kurzhals, L. Graf-Vlachy, and A. König, "Strategic leadership and technological innovation: A comprehensive review and research agenda," *Corp. Gov. An Int. Rev.*, vol. 28, no. 6, pp. 437–464, 2020, doi: 10.1111/corg.12351.

[79] J. Choi, N. Shin, and Y. S. Chang, "Strategic Investment Decisions for Emerging Technology Fields in the Health Care Sector Based on M&A Analysis," *Sustainability*, vol. 13, no. 7, 2021, doi: 10.3390/su13073644.

[80] T. N. Hai, Q. N. Van, and M. N. T. Tuyet, "Digital transformation: Opportunities and challenges for leaders in the emerging countries in response to covid-19 pandemic," *Emerg. Sci. J.*, vol. 5, no. Special Issue, pp. 21–36, 2021, doi: 10.28991/esj-2021-SPER-03.

[81] M. A. Arifin and F. P. Tajudeen, "Impact of human resources information systems in the military environment," *Asia Pacific Manag. Rev.*, vol. 25, no. 4, pp. 198–206, Dec. 2020, doi: 10.1016/J.APMRV.2020.02.001.

[82] X. Pan, S. Guo, M. Li, and J. Song, "The effect of technology infrastructure investment on technological innovation ——A study based on spatial durbin model," *Technovation*, vol. 107, p. 102315, Sep. 2021, doi: 10.1016/J.TECHNOVATION.2021.102315.

[83] R. Shi, M. Xu, and L. Zhu, "New techniques of remote sensing in the university of architecture and planning," *Int. Geosci. Remote Sens. Symp.*, vol. 2, p. 5418168, 2009, doi: 10.1109/IGARSS.2009.5418168.

[84] A. I. Prados *et al.*, "Impact of the ARSET program on use of remote-sensing data," *ISPRS Int. J. Geo-Information*, vol. 8, no. 6, p. 8060261, 2019, doi: 10.3390/ijgi8060261.

[85] S. Majumdar, "The Role of Remote Sensing and GIS in Military Strategy to Prevent Terror Attacks," *Intell. Data Anal. Terror Threat Predict. Archit. Methodol. Tech. Appl.*, pp. 79–94, 2021, doi: 10.1002/9781119711629.ch4.

[86] K. Lewis, "Challenges in military remote sensing," *Electro-Optical Remote Sensing, Detect. Photonic Technol. Their Appl.*, vol. 6739, p. 67390J, 2007, doi: 10.1117/12.753508.

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